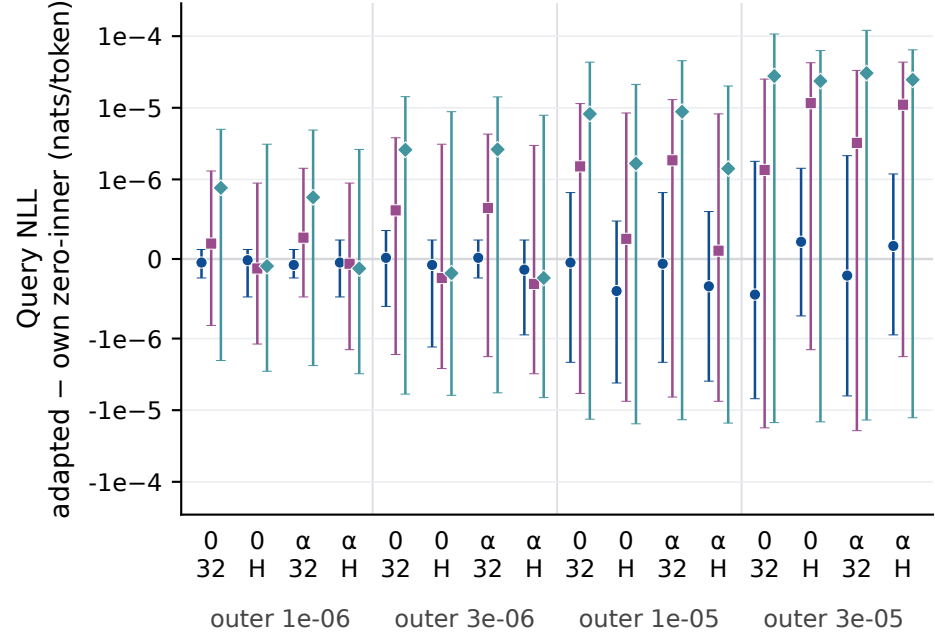


Local loss differences for three update maps

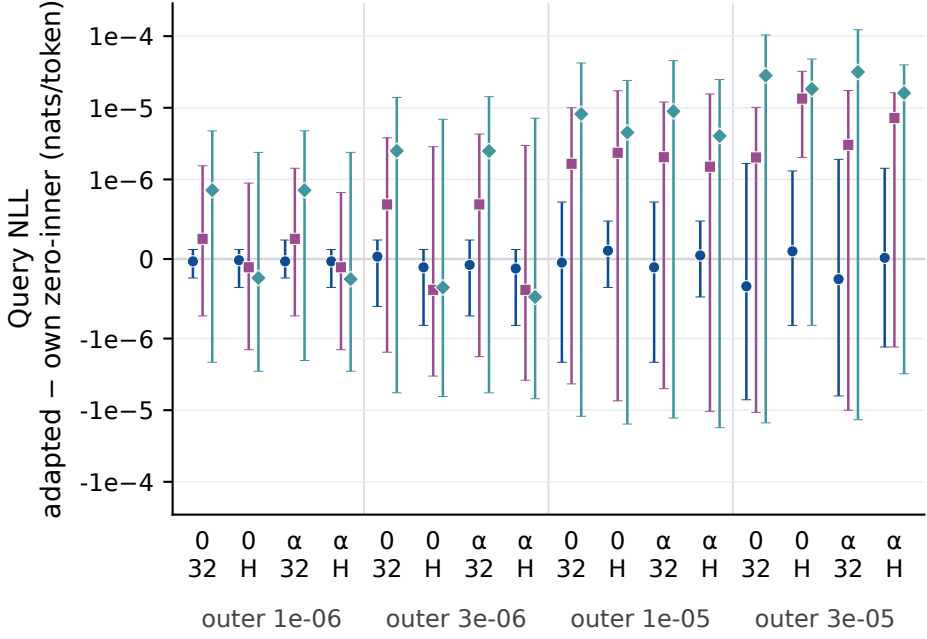
All 64 saved states × three paired maps · actual realized FP32 states · negative means lower local NLL

a Code · one pass



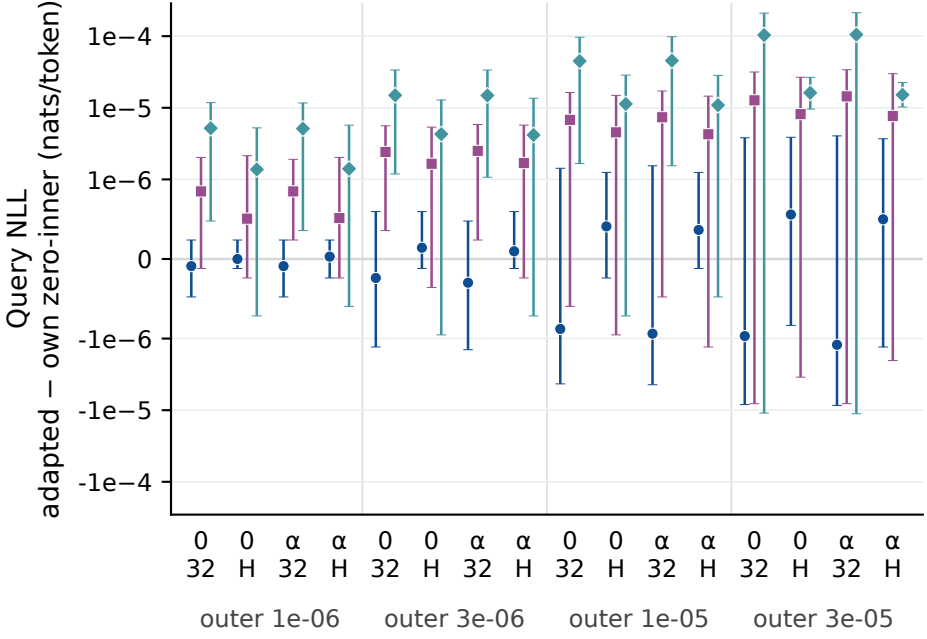
Ticks: history ($\alpha = 3e-5$) / saved step

b Code · fixed quarter, four passes



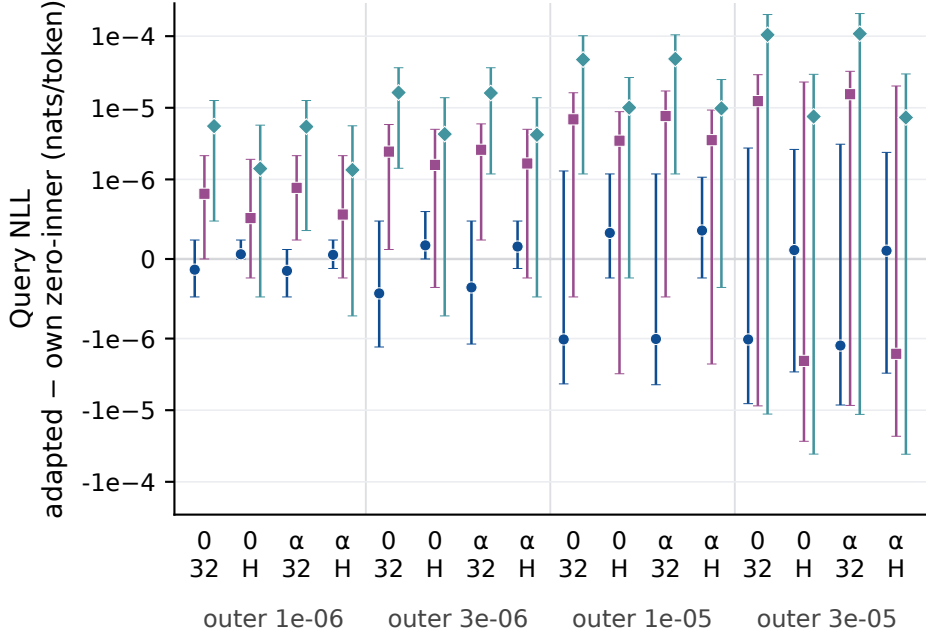
Ticks: history ($\alpha = 3e-5$) / saved step

c Math · one pass



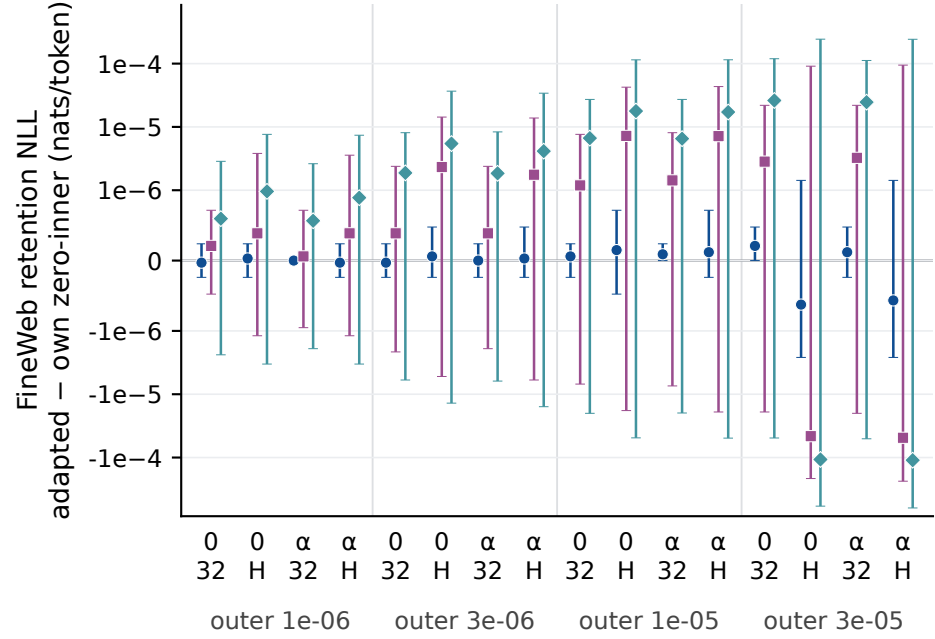
Ticks: history ($\alpha = 3e-5$) / saved step

d Math · fixed quarter, four passes



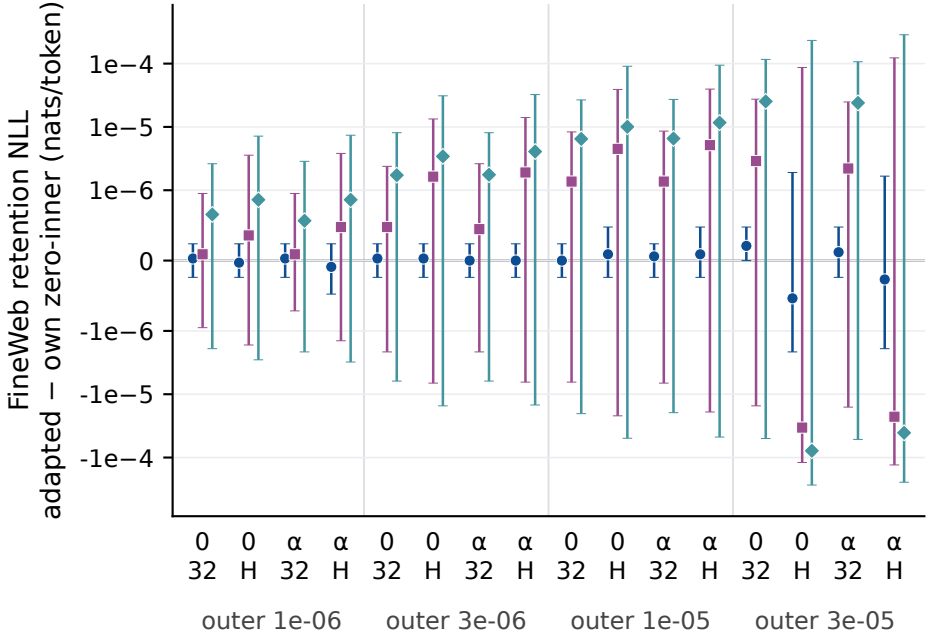
Ticks: history ($\alpha = 3e-5$) / saved step

e Code · one pass



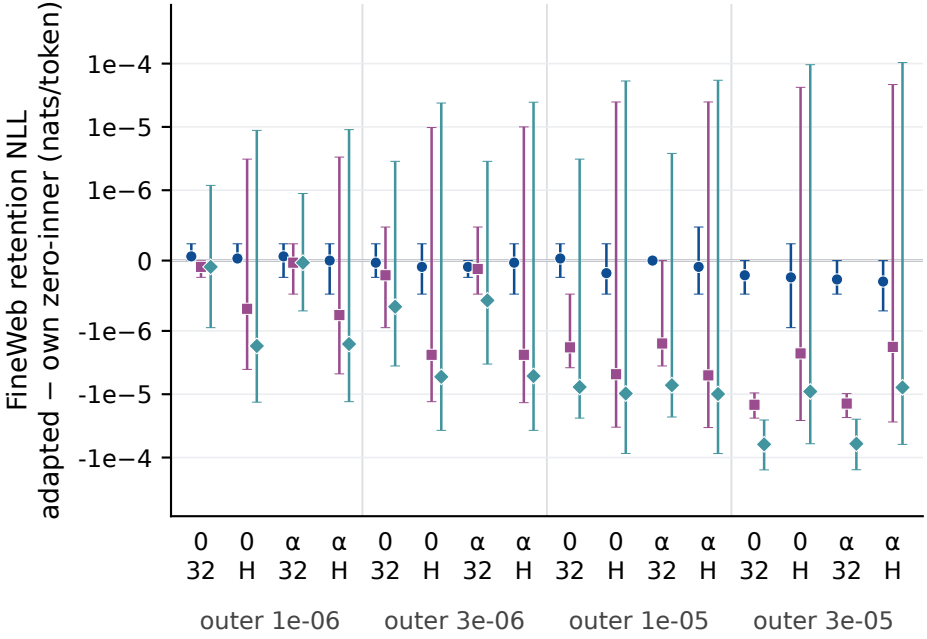
Ticks: history ($\alpha = 3e-5$) / saved step

f Code · fixed quarter, four passes



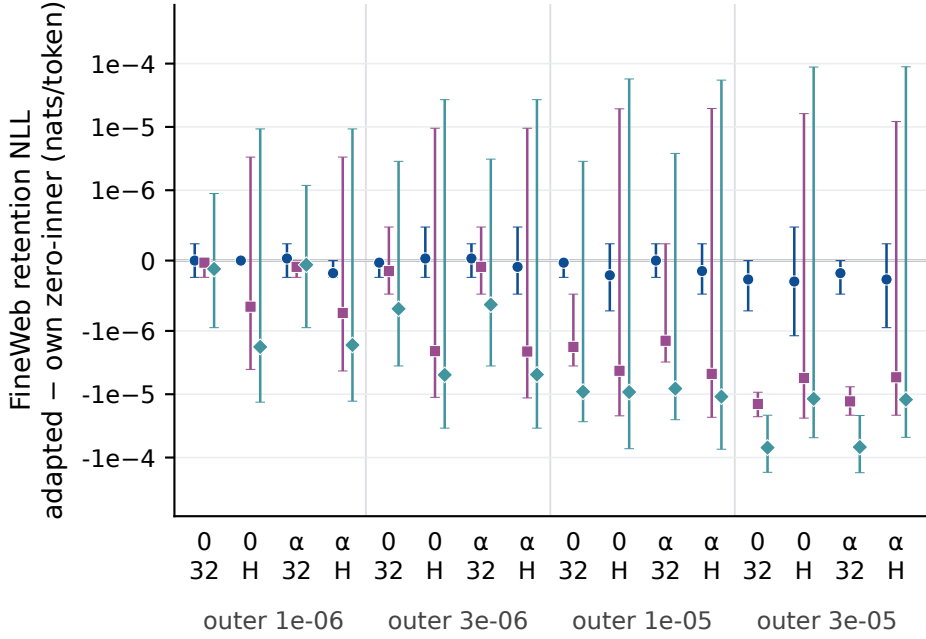
Ticks: history ($\alpha = 3e-5$) / saved step

g Math · one pass



Ticks: history ($\alpha = 3e-5$) / saved step

h Math · fixed quarter, four passes



Ticks: history ($\alpha = 3e-5$) / saved step

● Actual AdamW ■ Shared first moment; no adaptive scaling ◆ Clipped gradient; no momentum

Markers: means; whiskers: minimum-maximum of all eight fixed TRAIN triples. Signed symmetric-log axes retain zeros and adverse effects. Each map compares adapted to its own zero-inner arm. Alternatives match the zero Adam adaptive-component norm and use the same decay. Equal component norms need not give equal realized FP32 displacements. The separate norm-matched Adam vector has no loss evaluation. Step 32 uses the next scheduled update. H uses terminal moments for a hypothetical extra update at the last training rate. One seed and dependent local TRAIN diagnostics; ranges are not confidence intervals, p-values or held-out efficacy evidence.